Exploring Conflict-Matching Learning With Temporal Weight-Sharing and Bandwise Spatial-Interacting Transformer for Hyperspectral Change Detection

Yixiang Huang, Lifu Zhang, *Senior Member, IEEE*, Yi Cen, Ruoxi Song, Wenchao Qi, and Changping Huang

***Abstract*— Hyperspectral imagery is valuable for accurate detection of land-cover changes within a consistent area across time. However, current training paradigms for hyperspectral change detection (CD) are usually in supervised form which are not consistent to heavy labeling cost reality. Moreover, current deep learning methods face limitations due to insufficient temporal dependencies' sharing and inadequate densely bandwise spatial position dependencies. To tackle these challenges, we introduce an innovative semi-supervised training paradigm called Conflict HyperMatch and a deep learning model called temporal weight-sharing and bandwise spatial interacting transformer (TWBSIT) for hyperspectral CD. The key contributions of this study are as follows: 1) introduction of the Conflict HyperMatch training schedule, which relies on representation discrepancy and weak-to-strong prediction consistency, to improve the sample leveraging of both labeled and unlabeled data; 2) weight-sharing interacting temporal attention (WITA) module is contributed to capture shared temporal interactions and resemblances; and 3) bandwise cross-spatial attention (BCSA) module is introduced to achieve a more extensive spatial perception of the specific central image patch. Extensive experiments conducted on three real datasets validate the effectiveness of the proposed TWBSIT model based on Conflict HyperMatch in leveraging both labeled and unlabeled samples for hyperspectral CD. This method notably decreases the requirement for labeled training samples and surpasses the performance of many existing hyperspectral CD methods.**

***Index Terms*— Change detection (CD), deep learning, hyperspectral image, semi-supervised learning, transformer.**

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I. Introduction

CHANGE detection (CD) approaches in remote sensing images are designed to recognize land-cover distinctions within the same geographical region by examining images obtained at different time instances [1]. Various fields, including farmland monitoring [2], forest carbon evaluation [3], and ice shelf disintegration [4], have applied diverse CD methods to scrutinize the evolution of land cover. Hyperspectral remote sensing, a pivotal imaging technique in remote sensing [5], [6], [7], [8], is acknowledged for its ability to capture intricate variations in land-cover surfaces through an extensive array of spectral measurements, promoting CD techniques to uncover subtle change information. The process of hyperspectral CD encompasses three key stages: initial data preprocessing to enhance data quality, application of specialized CD algorithms to generate change maps, and performance evaluation to ensure accuracy and reliability of results.

Based on the methodology and the era of emergence, the hyperspectral CD methods can be categorized into two primary domains: traditional methods and deep-learning-based methods. In the realm of traditional hyperspectral CD, there are various subcategories, including methods based on image comparison operators for postclassification algorithm such as support vector machines (SVMs) [9], transformation-based approaches, and methods reliant on independent image classification. Image comparison operator-based methods such as change vector analysis (CVA) and distance measures are designed to detect changes by quantifying pixel discrepancies between the hyperspectral image pair [10], [11]. Nevertheless, these methods frequently face the intricate challenge of choosing an optimal threshold for accurately detecting changes. The transformation-based methods aim to reduce high-dimensional data and convert the original data into a smaller feature space, which inevitably leads to an inherent loss of information. Examples of such methods include principal component analysis (PCA) [12], [13] and band selection techniques [14], [15]. As for the independent image classification methods, the CD process relies on two distinct classification stages to produce a consolidated change map [16], [17]. This approach is vulnerable to

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error propagation from both the image classification stages, potentially affecting the accuracy of the detection results.

In recent years, the domain of hyperspectral CD has experienced a significant transformation, propelled by the integration of deep learning techniques, which notably enhances the change feature perceptions of CD methods. Among deep learning methods, transformer-based approaches do not gain popularity until earlier studies on convolutional networks revealed their weakness in capturing long-range dependencies. Therefore, deep leaning methods can furthermore be categorized into classical convolution-based model and transformer architecture-based model. Convolution-based methods usually leverage local features in the spectral and spatial domains to achieve improvements. For example, GETNET proposed LSConvolution to extract fused local spectral and spatial information [18]. But in most of the situations, convolution methods predominantly focus on feature extraction in a single spectral or spatial dimension, using 1-D [19], 2-D [20], [21] or 3-D convolutions [22], [23], which cannot jointly capture long-range temporal–spatial–spectral resemblance. For the other branch of deep learning methods, some transformer-based methods emerge in hyperspectral CD such as [24], [25], [26]. However, these methods still show drawbacks in temporal dependencies' sharing and cannot capture more densely bandwise spatial position dependencies in technical implementation.

Moreover, deep learning methods usually need a large number of training samples to encourage model to learn effective features. However, this basic requirement could not be satisfied in real situation of the hyperspectral domain. In other words, the labeling cost for training samples is too expensive for hyperspectral analysis [27], especially in hyperspectral CD. Combining the high-dimensional nature of hyperspectral data with the intricate temporal and spatial dynamics inherent in CD datasets, a shortage of training samples would be further exacerbated, impacting the final detection accuracy. Although there have been recent attempts to address this challenge through active learning approaches, they still face limitations due to the inevitable requirement of human participation [8], [28]. Consequently, these limitations continue to impact the performance of current deep learning models, restraining their ability to accurately detect changes in real-world scenarios.

To address these mentioned limitations of the existing methods, we introduce a novel semi-supervised approach for hyperspectral CD in this article: Conflict HyperMatch-based temporal weight-sharing and bandwise spatial interacting transformer (TWBSIT). Our proposed method leverages semi-supervised training strategy, and the training schedule is divided into two parts. The first step is to leverage both a small number of labeled samples and a large number of unlabeled samples to provide the model with a sufficient temporal–spatial–spectral feature extraction potential. For the unlabeled samples, we propose conflict loss function which combines discrepancy loss and consistency loss. This loss can encourage model to extract consistent multidimensional features while distinguish perturbations from views with different augmentations. As for labeled samples, they are trained for class-orientated convergence simultaneously. After the first step is finished, the parameters of TWBSIT are reserved and transferred to the second step which is the fine-tuning procedure. In the fine-tuning procedure, a small number of labeled samples, which are the same as that in the first step, are leveraged again to realize regular supervised learning. With reference to the core feature extraction model, TWBSIT consists of two main module contributions which are weight-sharing interacting temporal attention (WITA) module and bandwise cross-spatial attention (BCSA) module.

The contributions of this article are as follows.

1) *Conflict HyperMatch paradigm:* The proposed TWBSIT model is carried out in semi-supervised learning strategy, which is called the Conflict HyperMatch strategy. In Conflict HyperMatch, the TWBSIT is pretrained and initialized with both a small set of labeled samples and many unlabeled samples to address the challenge of heavy labeling cost. The unlabeled samples are augmented with weak and strong perturbations to create different views of the same input. Then through discrepancy loss and consistency loss, the TWBSIT can be facilitated in reasoning the same predictions while distinguishing representations that are from different views. At the same time, labeled samples synchronously allow the TWBSIT to extract classification-focused temporal–spatial–spectral resemblances.
2) *Shared temporal interaction capture:* To capture the shared temporal interaction information, the WITA module is contributed. This module enables the model to adaptively capture resemblance weighting between multitemporal images and share spectral and spatial evolution over time, which is crucial for promoting the flow of hyperspectral CD information across different temporal dimensions.
3) *Bandwise cross-spatial resemblance:* Long-range spatial resemblances are captured by the BCSA module. This component enhances the model's ability to achieve a wider spatial perception of the specific central image patch, which is due to the cross-spatial bandwise resemblances show different significance for change information detection and the spatial relationships are intensively computed.

II. Related Work

A. Semi-Supervised Learning

Semi-supervised learning has been extensively studied in vision tasks [29]. Among this semi-supervised learning strategy, based on weak-to-strong consistency, methods of match series turn to be more popular due to their effective unlabeled sample using [30], [31]. Since the heavy labor cost, which leads to insufficient training samples, is prevalent in hyperspectral analysis as well, the trend of using semi-supervised learning to provide sufficient training samples and extract features is becoming more popular [27]. Within the hyperspectral classification, semi-supervised training framework can be leveraged to tackle the issue of imbalanced data distribution by incorporation of spatial neighborhood

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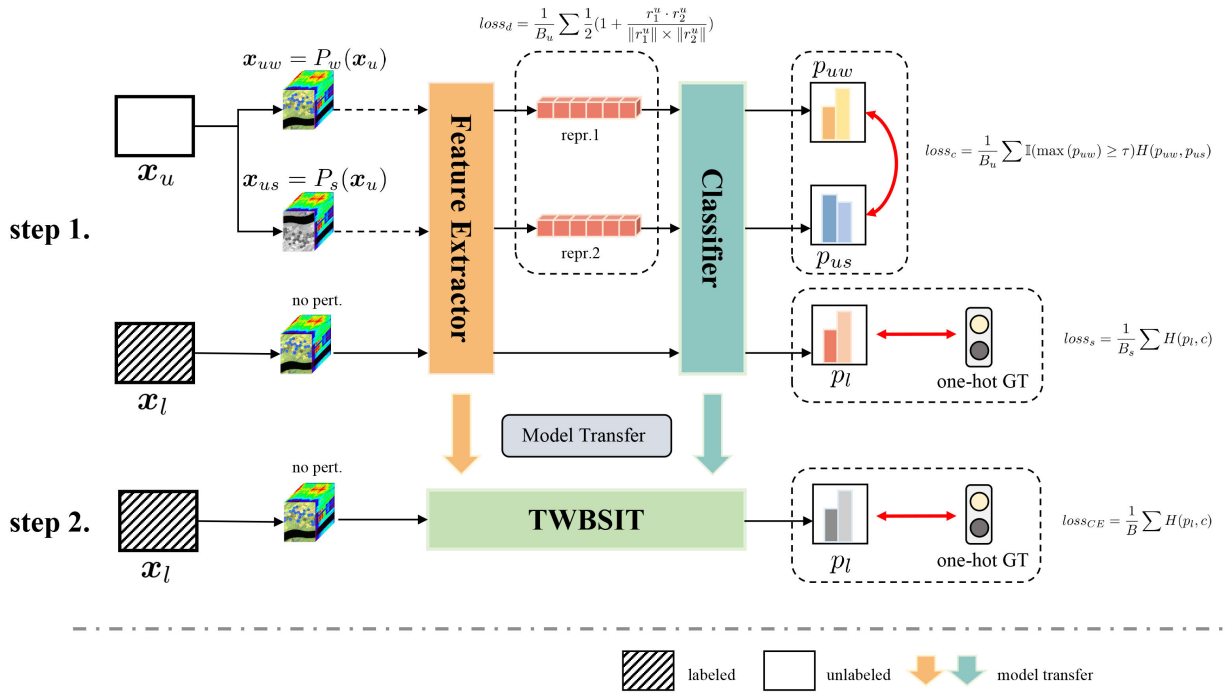


Fig. 1. Overall of the innovative semi-supervised paradigm Conflict HyperMatch. Step 1 consists of the discrepancy loss $\mathcal{L}_d$, the consistency loss $\mathcal{L}_c$, and the supervised loss $\mathcal{L}_s$ for both unlabeled and labeled samples. Step 2 is responsible for fine-tuning.

information [32]. Some other strategies, such as active learning [33], discriminative random field [34], and band selection [35], are integrated with the semi-supervised learning framework. To find profitable hyperspectral significant features, these methods can enhance the labeling process and improve the search for the most relevant spectral bands. In the hyperspectral CD domain, current semi-supervised training research is mostly implemented with pseudolabeling approaches. The pseudolabeling is applied for unlabeled samples which leverages CVA [36] or multilayer cascade screening [37] to generate an initial change map. The highly confident pseudosamples, along with real-labeled samples, can support the effective training of deep learning methods, effectively addressing the challenge posed by the limited availability of labeled data. Although these works try their best to leverage both labeled and unlabeled samples with various manners, the consistency of them is still implemented with strict class boundary, e.g., threshold. Therefore, the semi-supervised learning strategy in hyperspectral CD cannot figure out the deep change feature differences that are from different augmentation views, while preserving a consistent probability prediction.

B. Transformer Model

The transformer model was originally proposed for natural language processing and is equipped with an attention mechanism. It can effectively address sequence-to-sequence transformation problems by aggregating comprehensive contextual information from entire sequential inputs [38]. In the remote sensing domain, current AI techniques have developed very rapidly, and SpectralGPT has now become one of the most advanced research topics [46]. In hyperspectral image processing, the transformer framework can adaptively filter out irrelevant spectral and spatial details [39]. For classification tasks, rich spatial–spectral dependencies are extensively captured using the transformer framework [40], [41], whether independently or jointly. In hyperspectral CD, most methods are still focus on spatial and spectral information. For example, the attention module can use the resemblance of various spectral channels and spatial locations from multitemporal images to adaptively emphasize informative contributions of channels and locations [42]. To reduce influence to massive spectral and spatial information caused by patch size, adaptive spectral and spatial attention modules using Gaussian distributions are developed [43]. Some other methods adopt long-range dependencies on temporal domain, such as [24] and [25]. Although these methods have discussed the potential of leveraging temporal dependencies, the time-based transformer is still in the early stages. Specifically, the current transformer model for hyperspectral CD usually treats multitemporal images as a whole token set, which means that the shared resemblances from independent images cannot be effectively captured. Moreover, long-range spatial resemblances contribute differently from band to band, while they are often underused. Therefore, synthetically leveraging temporal–spatial–spectral information holds vital significance for the eventual CD effects.

III. Proposed Method

To address the mentioned challenges and improve the temporal–spatial information interacting in multitemporal hyperspectral images, we introduce the semi-supervised Conflict-HyperMatch-based TWBSIT for hyperspectral CD. The Conflict HyperMatch is responsible for fully leveraging

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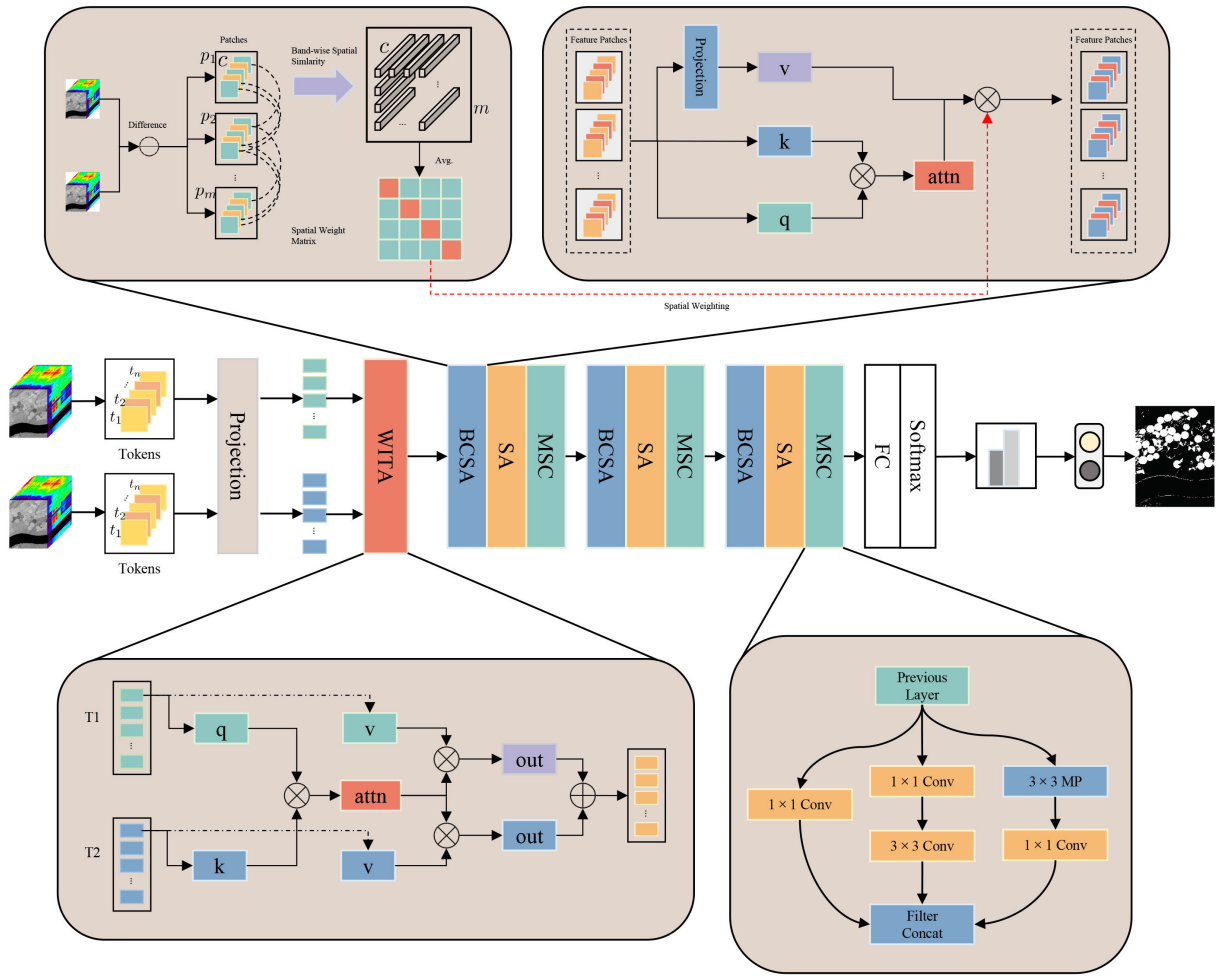


Fig. 2. Overall architecture of the proposed TWBSIT model. TWBSIT model mainly consists of WITA module and BCSA module.

information from both the labeled and unlabeled samples, using the image discrepancy concept and the weak-to-strong consistency principle. And the TWBSIT is account for adeptly integrating temporal and spatial interacting features. The training strategy of semi-supervised Conflict HyperMatch paradigm is illustrated in Fig. 1, and the overall model design of the innovative TWBSIT is shown in Fig. 2.

In this section, the proposed method is divided into four distinct sections, each offering in-depth explanation to specific aspects. The segments consist of Conflict HyperMatch learning strategy, the holistic design of the TWBSIT model, the WITA module, and the BCSA module.

A. Conflict HyperMatch

As depicted in Fig. 1, the Conflict HyperMatch is used in a two-step process to train the TWBSIT model. In the initial step, labeled data are used to enable the deep model to extract additional features from unlabeled samples. The second step involves fine-tuning the model transferred from the step 1 with labeled samples again.

In the Conflict HyperMatch strategy, the unlabeled and labeled samples are extracted with different procedure. Unlabeled samples $\boldsymbol{x}_u$ are perturbated with different series of augmentation operations, which are weak and strong perturbation. The weak perturbation $P_w(\cdot)$ contains random horizontal flips and random rotation operation to add minor adjustments on original hyperspectral patches, while the strong perturbation $P_s(\cdot)$ comprises random horizontal flips, random vertical flips, random rotation operations, and the Gaussian noise. These perturbations are sequentially organized. Follows show the perturbated versions $\boldsymbol{x}_{uw}$ and $\boldsymbol{x}_{us}$ of unlabeled samples:

$$\boldsymbol{x}_{uw} = P_w(\boldsymbol{x}_u) \tag{1}$$

$$\boldsymbol{x}_{us} = P_s(\boldsymbol{x}_u). \tag{2}$$

After that, the perturbated sample versions are sent into feature extractor and two representations in vector shape are outputted as $\boldsymbol{r}_1^u$ and $\boldsymbol{r}_2^u$. Here, we minimize the cosine similarity between two representations. This enables the feature extraction network to reason the inputs with different views, which is corresponding to the different perturbations. In other words, we need these two vector representations to show a far distance and more difference. The discrepancy loss between two representations can be formulated as follows:

$$\mathcal{L}_d = \frac{1}{B_u} \sum \frac{1}{2} \left(1 + \frac{\boldsymbol{r}_1^u \cdot \boldsymbol{r}_2^u}{\|\boldsymbol{r}_1^u\| \times \|\boldsymbol{r}_2^u\|} \right) \tag{3}$$

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where B_u indicates the batch size of unlabeled samples. Moreover, to ensure the discrepancy loss is always nonnegative and constrained within 0–1, the coefficients 1 and (1/2) are applied to correct the loss value.

Meanwhile, the representations are sent into a classifier to generate the probabilities to each class. Based on the weak-to-strong consistency, the probability $\boldsymbol{p}_{us}$ predicted from strongly perturbated samples should keep the consistent probability $\boldsymbol{p}_{uw}$ predicted from the weakly perturbated version. Furthermore, the implicit pseudosamples with high confidence can be filtered through both $\boldsymbol{p}_{uw}$ and the threshold τ. Therefore, the consistency loss for unlabeled samples can be defined as follows:

$$\mathcal{L}_c = \frac{1}{B_u} \sum \mathbb{I}(\max(\boldsymbol{p}_{uw}) \geq \tau) H(\boldsymbol{p}_{uw}, \boldsymbol{p}_{us}) \tag{4}$$

where the indicator function $\mathbb{I}(\cdot)$ is used to screen the predictions with high probabilities. $H(\cdot)$ indicates the entropy between two probabilities, which can be defined as follows:

$$H(\boldsymbol{p}_{us}, \boldsymbol{p}_{uw}) = -\sum_{i=1}^{c} \boldsymbol{p}_{us}^{i} \log\left(\boldsymbol{p}_{uw}^{i}\right). \tag{5}$$

Due to the aim of $\mathcal{L}_d$ is making two representations different, while the $\mathcal{L}_c$ is aimed to make two probabilities similar to each other. Therefore, the unsupervised loss for unlabeled samples $\mathcal{L}_u$ can be formulated as follows:

$$\mathcal{L}_u = \lambda_d \cdot \mathcal{L}_d + (1 - \lambda_d) \cdot \mathcal{L}_c \tag{6}$$

where λ_d is the balance coefficient.

In step 1, another training branch is arranged as a supervised manner. The labeled samples are sent into deep learning model without any perturbation. And the probability outputs $\boldsymbol{p}_l$ are combined with one-hot coded ground truth to calculate the cross-entropy loss. Therefore, the supervised loss $\mathcal{L}_s$ in step 1 can be represented as follows:

$$\mathcal{L}_s = \frac{1}{B_s} \sum H(\boldsymbol{p}_l, c) \tag{7}$$

where B_s denotes the batch size for labeled samples in step 1 and c indicates the ground truth. Finally, the loss for combination of $\mathcal{L}_s$ and $\mathcal{L}_u$ can be formulated as follows:

$$\mathcal{L} = \mathcal{L}_s + \lambda_u \cdot \mathcal{L}_u. \tag{8}$$

The λ_u is the coefficient to adjust the influence from $\mathcal{L}_u$. After the training schedule of step 1 being finished, the trained model, both feature extractor and classified, is transferred to step 2.

In step 2, the unlabeled samples are excluded from training samples, and only the limited labeled samples are used to fine-tune the transferred model. With a different batch size from step 1, the fine-tuning cross-entropy loss function is defined as follows:

$$\mathcal{L}_{\mathrm{CE}} = \frac{1}{B} \sum H(\boldsymbol{p}_l, c). \tag{9}$$

Note that the values of parameter B and parameter B_s do not influence each other.

B. TWBSIT

The overall design for TWBSIT is shown in Fig. 2. In our proposed Conflict HyperMatch framework, the TWBSIT plays a vital role in extracting temporal, spatial, and spectral interaction. The TWBSIT mainly consists of WITA module, BCSA module, spectral attention (SA) module, and multiscale convolution (MSC).

As Fig. 2 illustrates, in TWBSIT, the multitemporal hyperspectral images are organized into token-based patches and these original n tokens which are $T^i = \{t_1^i, t_2^i, \ldots, t_n^i\}, i = 0, 1$, are regularized and projected into tokens in length of l, where i indicates the index of time. Then the projected tokens are sent into WITA to capture shared temporal interactions. The procedure can be described an follows:

$$T_P^i = F_P(T^i) \tag{10}$$

$$\boldsymbol{x}_{\mathrm{WITA}} = F_R\left(F_{\mathrm{WITA}}\left(T_P^0, T_P^1\right)\right) \in \mathbf{R}^{c \times h \times w}. \tag{11}$$

Here, $F_R(\cdot)$ indicates the reshape operation and $F_{\mathrm{WITA}}(\cdot)$ denotes the feature extraction in the WITA module. T_P^0 and T_P^1 are the projection for input tokens which are also composed of a series of n elements, respectively.

Subsequently, a sequence of module stacks is settled to capture cross-spatial resemblance with bandwise spatial similarity, inherent spectral dependencies, and multiscale spatial features. Among the stack of modules, the detail design of BCSA is arranged into Section III-B2, and the SA module is derived from our past work [7]. As for the MSC, it is proposed to focus on different spatial information extraction in multiscale receptive fields. To obtain an optimal local sparse structure and make model accommodative to multitemporal hyperspectral image patches, the MSC is adjusted and improved with three branches of feature extraction

$$\begin{aligned}\boldsymbol{x}_{\mathrm{MSC}}^{\mathrm{out}} = F_{\mathrm{Concat}}\big(&F_{\mathrm{Conv}}^{1\times1}\left(\boldsymbol{x}_{\mathrm{MSC}}^{\mathrm{in}}\right), F_{\mathrm{Conv}}^{3\times3}\left(F_{\mathrm{Conv}}^{1\times1}\left(\boldsymbol{x}_{\mathrm{MSC}}^{\mathrm{in}}\right)\right)\\ &F_{\mathrm{Conv}}^{1\times1}\left(F_{\mathrm{MP}}^{3\times3}\left(\boldsymbol{x}_{\mathrm{MSC}}^{\mathrm{in}}\right)\right)\big)\end{aligned} \tag{12}$$

where $F_{\mathrm{Conv}}^{1\times1}(\cdot)$ and $F_{\mathrm{Conv}}^{3\times3}(\cdot)$ indicate the convolution layers. $F_{\mathrm{MP}}^{3\times3}(\cdot)$ denotes the max pooling layer. After temporal spatial and spectral interaction features being extracted, the feature maps are converted into probabilities to specific classes through classifier, and the change map can be generated.

1) WITA: For the change feature interaction in the temporal domain, the WITA module is proposed to capture shared resemblance for multitemporal images. Nowadays, temporal interaction has been discussed through deep-learning-based methods and has achieved good results. However, most current deep learning methods concentrate the logical sequence temporal change interaction, which ignored the fact that change resemblances are equally important to both the temporal images. In this article, we explore the shared correlation and realize the change information interaction for each multitemporal image, respectively. The architecture of the WITA module is illustrated in Fig. 3.

To realize the cross-temporal interaction and keep independent images share the same correlation weights, the WITA module is arranged as follows. The first temporal image is regarded as the query vectors, and the second temporal image

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Fig. 3. Architecture of the WITA module.

is arranged as key vectors. Since the feature vector is divided into tokens in length of lhw, the corresponding attention maps can be formulated as follows:

$$\boldsymbol{am}_{\text{share}} = F_{\text{SoftMax}}\left(\frac{\boldsymbol{q} \otimes \boldsymbol{k}}{\sqrt{d_k}}\right) \in \mathbf{R}^{n \times n} \tag{13}$$

where $\boldsymbol{q} \in \mathbf{R}^{n \times lhw}$ and $\boldsymbol{k} \in \mathbf{R}^{lhw \times n}$ indicate the corresponding tokens which are acquired from the first temporal T_0 and second temporal T_1 image patches, respectively. The mark of $\otimes$ indicates the multiplication of matrices. $F_{\text{SoftMax}}(\cdot)$ is represented for the activate function of SoftMax. Following the original purpose of numerical stability, the parameter d_k represents the dimensions of queries and keys, serving as a scaling factor. Since the shared attention maps are calculated from both the temporal image patches, the attention maps should be weighted with value vectors which are from T_0 and T_1, respectively,

$$\boldsymbol{out}_0 = \boldsymbol{am}_{\text{share}} \otimes \boldsymbol{v}_0 \in \mathbf{R}^{n \times lhw} \tag{14}$$

$$\boldsymbol{out}_1 = \boldsymbol{am}_{\text{share}} \otimes \boldsymbol{v}_1 \in \mathbf{R}^{n \times lhw} \tag{15}$$

where $\boldsymbol{v}_0$ and $\boldsymbol{v}_1$ are tokens which are copies from the corresponding temporal image patches. It is worthy to note that shared attention maps originated from two different temporal patch inputs facilitate cross-temporal interaction for both the separated temporal patches again. Finally, the transition layer composed of concatenation and convolution layer is settled to fuse temporal features, which can be expressed as follows:

$$\boldsymbol{out} = F_{\text{Conv}}^{1\times1}(F_R(F_{\text{Concat}}(\boldsymbol{out}_0, \boldsymbol{out}_1))) \in \mathbf{R}^{c \times h \times w}. \tag{16}$$

2) BCSA: Spatial dependencies play a vital role in finding boundary change information and improving model perception of entire data structure. There is some research making spatial attention mechanism work on hyperspectral image analysis. However, these works only concentrate the inner spatial correlations within a patch size and do not leverage more dense and long-range spatial interaction between each band pairs. To facilitate use of intensive bandwise spatial correlations, we propose the BCSA module. The architecture design of BCSA is illustrated in Fig. 4.

The process for BCSA can be divided into two parts: computation of bandwise spatial similarity and cross-spatial attention mechanism. During the computation of bandwise spatial similarity, the difference map is initially acquired from the multitemporal hyperspectral image pairs and then using stratified random sampling to generate a batch of m image patches. Then the bandwise cosine similarities are calculated between each patch. The cosine similarity cube of band pairs can be expressed in formula

$$S_{i,j}^k = \frac{\boldsymbol{x}_i^k \cdot \boldsymbol{x}_j^k}{\|\boldsymbol{x}_i^k\| \|\boldsymbol{x}_j^k\|}, \quad i, j = 1, 2, \ldots, m, \ k = 1, 2, \ldots, c \tag{17}$$

where $\boldsymbol{x}_i^k$ and $\boldsymbol{x}_j^k$ are the kth band vector representations of the ith and jth patches, respectively. Since the cosine similarity value is calculated between the corresponding bands of patches, the final similarity matrix $S \in \mathbf{R}^{m \times m \times c}$ can be obtained. However, the similarity matrix S is still represented for the band-level relationships, which cannot express the whole spatial relationships for later weighting in cross-spatial interaction. The weighting matrix is averaged through the dimension of spectral bands and can be represented in formula

$$W = \frac{1}{c}\sum_{k=1}^{c} S^k \in \mathbf{R}^{m \times m} \tag{18}$$

where $S^k \in \mathbf{R}^{m \times m}$ is the kth similarity matrix in the spectral dimension.

The other branch of BCSA is the long-range cross-spatial interaction which is illustrated on the right side of Fig. 4. The original patches are simultaneously processed by the previous feature extraction layers except the similarity weighting matrix computation. After the feature extraction layers, the feature maps are arranged and operated in a scale of batch size group, as well. To obtain the cross-spatial resemblances, the group of patches are arranged as queries $Q \in \mathbf{R}^{m \times hw \times c}$ and keys $K \in \mathbf{R}^{m \times c \times hw}$ to adaptively calculate the attention weighting maps. The intensive attention maps for the ith query vector can be expressed as follows:

$$\boldsymbol{attn}_i = F_{\text{SoftMax}}\left(\frac{\boldsymbol{Q}_i \otimes \boldsymbol{K}}{\sqrt{d_k}}\right) \in \mathbf{R}^{m \times hw \times hw}, \ i = 1, 2, \ldots, m \tag{19}$$

where $\boldsymbol{Q}_i \in \mathbf{R}^{hw \times c}$ indicates the hw queries in the ith patch. $\boldsymbol{K}$ are all the key vectors for each patch. Therefore, the ith attention map for the corresponding queries is the production set with keys in each patch. Since the production is repeated for m times, the attention maps $\boldsymbol{attn} \in \mathbf{R}^{m \times m \times hw \times hw}$ can be obtained. For the final value weighting, we project the previous feature map to a new feature space for values $\boldsymbol{V} \in \mathbf{R}^{m \times hw \times c}$ with convolution. The value weightings for

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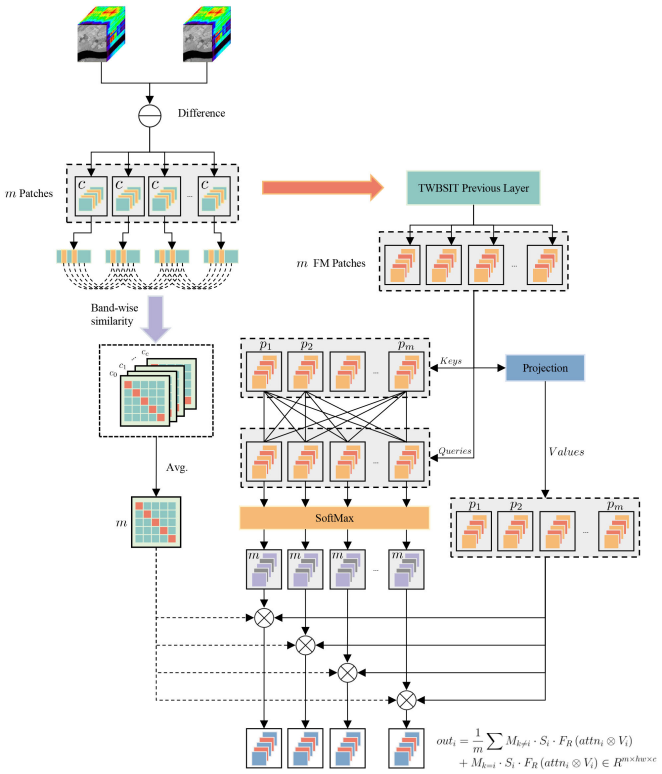


Fig. 4. Architecture design of the BCSA module.

each patch p_i are processed with both the similarity matrix and corresponding attention maps. Through intensive weighting for cross-spatial interaction, the model can capture more long-range resemblances in a large number of spatial features, and the output feature map ith patch can be expressed as follows:

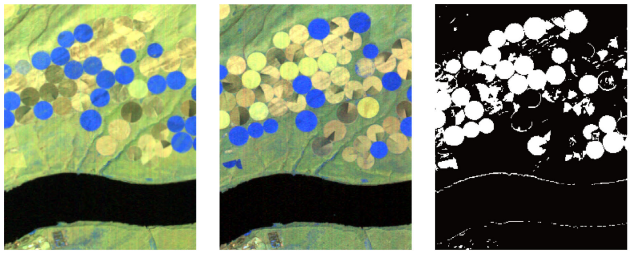
$$\boldsymbol{out}_i = \frac{1}{m}\sum M_{k\neq i}\cdot S_i \cdot F_R(\boldsymbol{attn}_i \otimes \boldsymbol{V}_i) + M_{k=i}\cdot S_i \cdot F_R(\boldsymbol{attn}_i \otimes \boldsymbol{V}_i) \in \mathbf{R}^{m\times hw\times c} \quad (20)$$

where M indicates the mask in the same shape with S_i. As the formula shows, we do not simply use linear combination and do not treat all the cross-spatial interactions equally, either. Instead, we treat the influence from other patches as a population and leverage their average to improve the perception to external spatial environment of the corresponding central patches.

IV. Experiments

A. Datasets

In this article, the performance of TWBSIT based on the Conflict HyperMatch strategy is evaluated across three publicly available multitemporal hyperspectral CD datasets. These datasets were acquired using the Earth Observing-1 (EO-1) satellite equipped with a Hyperion sensor, offering a spectral range of 0.4–2.5 μm with approximately 242 spectral bands and a spectral resolution of around 10 nm. The spatial resolution is 30 m. In the experiments, spectral bands with low signal-to-noise ratios (SNR) were excluded and binary ground-truth maps were generated through visual analysis and on-site inspections. The first dataset, the Irrigated Agricultural



(a) (b) (c)

Fig. 5. Irrigated agricultural dataset with false-color map (bands 134, 90, and 75 as RGB). (a) USA farmland image captured on May 1st, 2004. (b) USA farmland image captured on May 8th, 2007. (c) Binary ground truth for Irrigated Agricultural dataset.

dataset, depicts an irrigated agricultural area in Hermiston city, Umatilla County, Oregon, USA, with images captured on May 1st, 2004, and May 8th, 2007. After removing noise and bands with no data, it consists of 307 × 241 pixels and 156 bands [44]. The training samples follow their original configuration, comprising a total of 7250 pixels, which account for around 9.7%. Note that this training rate is serving as the benchmark rate for training sample analysis. The second dataset, the Wetland Agricultural dataset, represents a farmland area in Yuncheng, Zhejiang, China. It comprises images taken on May 3rd, 2006, and April 23rd, 2007, with a resolution of 450 × 140 pixels and 156 bands, following noise removal [44]. Because the first and second datasets basically contain similar land-cover type, we set the same training rate of 9.7% which is serving as the benchmark training rate, as well, in the Wetland Agricultural dataset to keep consistency. The final dataset, the River dataset, portrays a river area in Jiangsu, China, with images captured on May 3rd, 2013, and December 31st, 2013. It encompasses 463 × 241 pixels and 198 bands after noise removal [18]. Considering class imbalance issues in the River dataset, the proportion of unchanged samples to changed samples is maintained at 2:1, as in the original work. This results in a training rate of 4.03%, which serves as the benchmark for sample size analysis. The false-color and ground-truth maps of three datasets are shown in Figs. 5–7. The stratified random sampling and multiple experiments (five times) are applied to make comprehensive comparisons. The details of every dataset are shown in Table I.

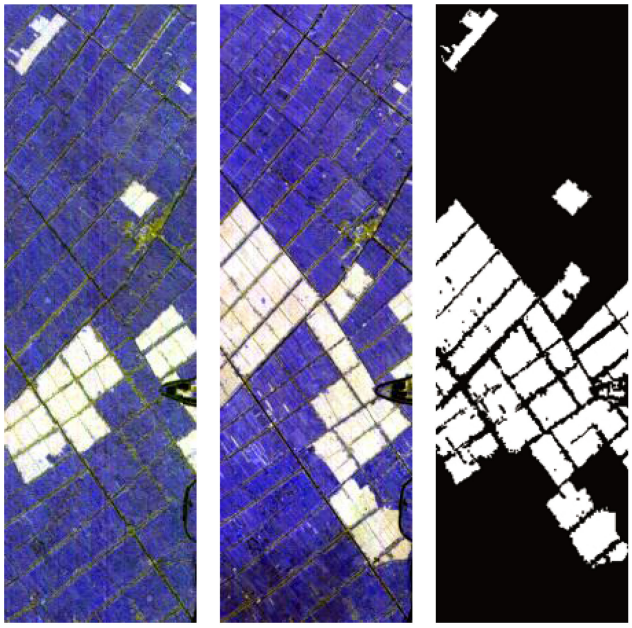
B. Competing Methods and Experimental Settings

To demonstrate the superior effectiveness of our proposed approach, we compare it against various conventional hyperspectral CD techniques and cutting-edge deep learning methods. One of the traditional methods under consideration is CVA (2007) [11], which uses the difference map of dual-temporal hyperspectral images and uses an unsupervised threshold method for change map segmentation. Another traditional approach involves SVM, a widely used method for land-cover CD (2006) [9]. Among the deep learning methods, we evaluate our model against GETNET (2019), which is based on the LSConvolution architecture [18]. Given that our approach incorporates spectral and spatial domain learning, we also include WCRN (2022) as a

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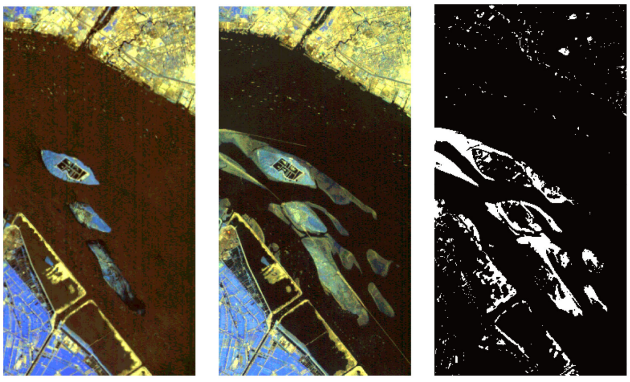
TABLE I
DETAIL OF THREE PUBLIC DATASETS

Dataset	Spatial Size	Band	Date1	Date2	Training Rate	Training Samples
Irrigated	307 × 241	156	May 1st, 2004	May 8th, 2007	9.7%	7250
Wetland	450 × 140	156	May 3rd, 2006	Apr. 23rd, 2007	9.7%	6173
River	463 × 241	198	May 3rd, 2013	Dec. 31st, 2013	4.03%	4500



(a) (b) (c)

Fig. 6. Wetland Agricultural dataset with false-color map (bands 134, 90, and 75 as RGB). (a) China farmland captured on May 3rd, 2006. (b) China farmland captured on April 23rd, 2007. (c) Binary ground truth for Wetland Agricultural dataset.



(a) (b) (c)

Fig. 7. River dataset with false-color map (bands 134, 90, and 75 as RGB). (a) China river captured on May 3rd, 2013. (b) China river captured on December 31st, 2013. (c) Binary ground truth for River dataset.

two-domain learning comparison method, implemented with a CNN architecture [21]. For transformer-based methods, we consider ChangeFormer (2023) [45] and SSTFormer (2023) [24]. In addition, our previous work TFR (2023), which integrates attention mechanism-based spatial–spectral module, is included in our comparisons as well [39]. We use the default parameter settings introduced in the corresponding works for these deep-learning-based methods.

All the experiments take place on an NVIDIA RTX 3060 with 12 GB of GPU memory for training and testing, and implementations of algorithms were run on the PyTorch platform. Due to the model's architecture, it accommodates various patch size inputs without imposing any specific spatial resolution restrictions. The Adagrad optimizer with 1×10^{-4} learning rate is used throughout the entire process, and we choose the cross-entropy loss function for calculating the supervised loss.

C. Evaluation Metrics

To ensure a fair and thorough performance comparison across the three hyperspectral CD datasets, we use various metrics, including overall accuracy (OA), changed accuracy (CA), kappa coefficient (Kappa), and $F1$-score. Higher values for these metrics indicate better model performance.

1) Overall Accuracy: In tasks resembling pixel-level classification, accuracy is a simple and effective metric to assess model performance. The OA is calculated using the following formula:

$$\mathrm{OA} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \tag{21}$$

2) Changed Accuracy: We use CA to assess the detection of changed areas, which is also known as precision. CA represents the proportion of true positives in the combined total of true positives and false positives. The formula is as follows:

$$\mathrm{CA} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FP}}. \tag{22}$$

3) Kappa Coefficient: Kappa is another metric commonly used for pixel-level classification. It accounts for class imbalances and provides a fair assessment of model performance across different datasets. The formula for calculating kappa is as follows:

$$\mathrm{Kappa} = \frac{\mathrm{OA} - \mathrm{PE}}{1 - \mathrm{PE}} \tag{23}$$

$$\mathrm{PE} = \frac{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN}) + (\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}{(\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN})^2}. \tag{24}$$

4) F1-Score: The $F1$-score is a composite metric used in binary classification evaluation, balancing the precision and recall of a model. It calculates effectiveness by considering true positives, false positives, and false negatives, offering a balanced assessment of the model's performance. $F1$-score values range from 0 to 1, where higher values indicate better model performance. It is particularly valuable for imbalanced datasets

$$F1\text{-Score} = \frac{2 \times \mathrm{CA} \times \mathrm{RE}}{\mathrm{CA} + \mathrm{RE}} \tag{25}$$

$$\mathrm{RE} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}}. \tag{26}$$

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TABLE II
MODEL COMPARISON RESULTS ON IRRIGATED AGRICULTURAL DATASET (REPEAT FIVE TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9286	0.7704	0.8127	0.8127
SVM	0.9462±0.0094	0.8233±0.0305	0.8558±0.0233	0.8911±0.0170
GETNET	0.9496±0.0031	0.8510±0.0133	0.8610±0.0079	0.8939±0.0058
WCRN	0.9420±0.0031	0.8308±0.0119	0.8412±0.0076	0.8790±0.0055
ChangeFormer	0.8736±0.0170	0.7874±0.0756	0.6080±0.0863	0.6831±0.0854
TFR-PS^{2}ANet	0.9709±0.0013	0.9217±0.0035	0.9181±0.0005	0.9368±0.0026
SSTFormer	0.9601±0.0054	0.8850±0.0325	0.8892±0.0134	0.9151±0.0098
No CHM	0.9787±0.0006	0.9421±0.0049	0.9396±0.0017	0.9513±0.0055
Proposed	**0.9801±0.0006**	**0.9541±0.0046**	**0.9433±0.0021**	**0.9561±0.0017**

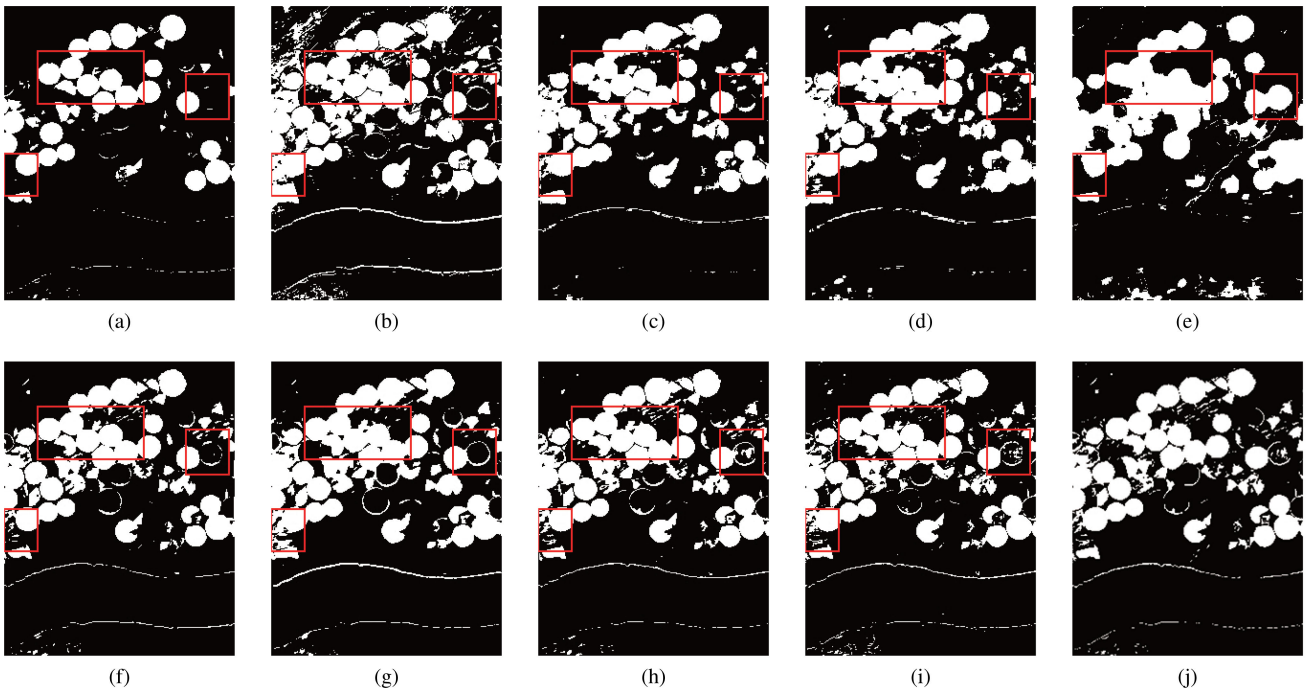


Fig. 8. Change map results of different methods on the Irrigated Agricultural dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) Without CHM. (i) Proposed CHM-based TWBSIT. (j) Ground truth.

D. Comparison Experiments

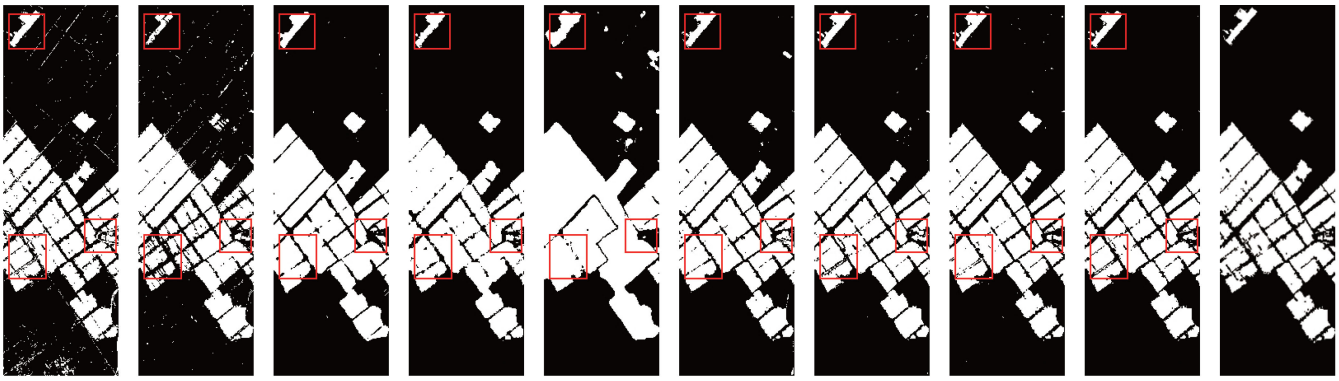
1) Experiments on the Irrigated Agricultural Dataset: The model comparison results for the Irrigated Agricultural dataset are presented in Table II. Our proposed Conflict-HyperMatch-based TWBSIT stands out among all the comparison methods, achieving superior performance on every evaluation metric, including OA, CA, Kappa, and $F1$-score. The results of TWBSIT without Conflict HyperMatch (No CHM) show inferior OA performance, indicating that the semi-supervised learning-based Conflict HyperMatch can effectively leverage unlabeled samples compared with the supervised framework in the standard training sample benchmark. The most competitive method is the TFR model which achieves OA of 0.9709, CA of 0.9217, Kappa of 0.9181, and $F1$-score of 0.9368. The second-best belongs to SST-Former, with an OA of 0.9601, CA of 0.8850, and Kappa of 0.8892, which is inferior to our proposed approach by approximately 0.02 OA. For the convolution-based methods of GETNET and WCRN, they show similar results of OA which are 0.9496 and 0.9420, respectively. The conventional SVM achieves an OA of 0.9462 and a CA of 0.8233. The other conventional CVA method shows a slight decrease, compared with the SVM method, which achieves OA of 0.9286 eventually. As a transformer architecture-based method, ChangeFormer achieves the worst performance of OA, Kappa, and $F1$-score, despite being a transformer-based deep learning method. This may be attributed to its inability to effectively handle complex and rich spectral information.

The CD result comparison maps for the Irrigated Agricultural dataset are depicted in Fig. 8. The Conflict-HyperMatch-based TWBSIT demonstrates the best visually similar results when compared with the ground-truth map. It effectively distinguishes change areas of both agricultural land and river border line. The main result differences are marked in every subfigure. The change results obtained by CVA show failure of detecting farmland areas and the borderline changes in the river also omitted. When it comes to the results obtained by SVM, the results seem very messy and mistakenly detect change areas in the top-left corner. The change results of GETNET and WCRN are similar to each other. Both the models lose much detailed border information of agricultural land. The ChangeFormer model shows the worst visual change map results due to every agricultural land is connected to each other and much subtle information is lost. The models of TFR and SST-Former are the competitive models with only little detail information lost within the round-shaped agricultural lands. The visual results

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TABLE III
MODEL COMPARISON RESULTS ON WETLAND AGRICULTURAL DATASET (REPEAT FIVE TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9525	0.9032	0.8859	0.9196
SVM	0.9497±0.0337	0.9088±0.0130	0.8748±0.0917	0.9092±0.0706
GETNET	0.9611±0.0037	0.8954±0.0133	0.9082±0.0082	0.9360±0.0054
WCRN	0.9602±0.0008	0.9045±0.0062	0.9054±0.0017	0.9337±0.0011
ChangeFormer	0.8636±0.0443	0.7646±0.0621	0.6651±0.1157	0.7597±0.0864
TFR-PS^{2}ANet	0.9804±0.0011	0.9630±0.0068	0.9526±0.0028	0.9664±0.0020
SSTFormer	0.9743±0.0033	0.9360±0.0051	0.9384±0.0098	0.9566±0.0067
No CHM	0.9788±0.0008	0.9480±0.0049	0.9491±0.0019	0.9641±0.0013
Proposed	**0.9812±0.0009**	**0.9581±0.0092**	**0.9548±0.0022**	**0.9680±0.0015**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 9. Change map results of different methods on the Wetland Agricultural dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) Without CHM. (i) Proposed CHM-based TWBSIT. (j) Ground truth.

of TWBSIT without Conflict HyperMatch prove that the proposed semi-supervised framework can effectively leverage unlabeled samples.

2) Experiments on the Wetland Agricultural Dataset: The model comparison results for the Wetland Agricultural dataset are outlined in Table III. In this dataset, the TWBSIT model based on Conflict HyperMatch surpasses other comparison methods across multiple metrics, such as OA, CA, Kappa, and $F1$-score. Even when excluding Conflict HyperMatch, our TWBSIT still exhibits notable performance in the comparison results. The best competitor belongs to TFR which shows a slight decrease in OA (around 0.006) compared with the Conflict-HyperMatch-based TWBSIT. The second-best competitor is SST-Former which shows a similar OA performance to TFR. GETNET achieves an acceptive performance with an OA of 0.9611, albeit showing a notable decrease (around 0.02) compared with the proposed method. CVA and SVM produce similar OA results, with values of 0.9525 and 0.9497, respectively. The two-domain learning WCRN demonstrates inferior OA and Kappa results, scoring 0.9602 and 0.9054, respectively. However, ChangeFormer's performance decreases most, resulting in an OA of 0.8636, and it exhibits the highest standard deviation among the comparison methods. This suggests its initial suitability for multispectral CD is limited effectiveness in handling complex hyperspectral images.

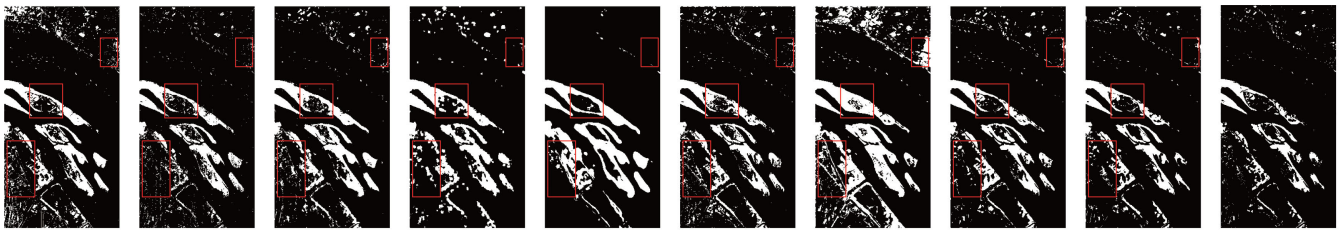
The change maps resulting from different methods for the Wetland Agricultural dataset are depicted in Fig. 9. The Conflict-HyperMatch-based TWBSIT produces change maps with visually similar effects to the ground-truth map. The predicted change map can effectively distinguish subtle changes in the farmland blocks and the unchanged areas are clear, as well. The main differences in every change map are marked with red box. The conventional approaches, CVA and SVM, inaccurately detect numerous borders between farmland blocks, displaying a considerable amount of noise-like detections in the left corner of the CD map. For convolutional deep learning methods, GETNET yields a change map that generally aligns with the ground truth, but it fails to capture the subtle changes within the farmland blocks. ChangeFormer and the two-domain WCRN demonstrate similar visual effects. However, they fundamentally struggle to capture any borderline change information within the farmland area, rendering them ineffective in detecting subtle changes. TFR, SST-Former, and TWBSIT without Conflict HyperMatch exhibit the closest visual resemblance to the ground-truth map among the comparison methods, preserving subtle change information. Nevertheless, excessive noise-like detections in the top-left portion impede the overall visual clarity.

3) Experiments on the River Dataset: The comparison results for different methods on the River dataset are presented in Table IV. The Conflict-HyperMatch-based TWBSIT we propose outperforms all other methods in terms of OA, CA, Kappa, and $F1$-score. When we remove the semi-supervised learning framework, the proposed TWBSIT can still achieve the best OA performance under standard training rate. Among comparison models, TFR is the best competitor to our approach, achieving an OA of 0.9488, CA of 0.6459, and a kappa index of 0.7340. The second-best performer is GETNET, achieving an OA of 0.9352, proving its effectiveness in datasets characterized by linear spectral relationships and local spatial texture information.

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TABLE IV
MODEL COMPARISON RESULTS ON RIVER DATASET (REPEAT FIVE TIMES)

Models	OA	CA	Kappa	F1-Score
CVA	0.9280	0.5492	0.6617	0.6992
SVM	0.9147±0.0393	0.5431±0.1098	0.5838±0.0998	0.6275±0.0844
GETNET	0.9352±0.0099	0.5829±0.0424	0.6821±0.0339	0.7163±0.0292
WCRN	0.9310±0.0061	0.5648±0.0249	0.6637±0.0181	0.7000±0.0154
ChangeFormer	0.9153±0.0189	0.5287±0.0908	0.5519±0.0523	0.5971±0.0430
TFR-PS^{2}ANet	0.9488±0.0084	0.6450±0.0477	0.7340±0.0336	0.7615±0.0293
SSTFormer	0.9061±0.0297	0.4955±0.0769	0.5929±0.0731	0.6396±0.0613
No CHM	0.9552±0.0045	0.6721±0.0240	0.7632±0.0178	0.7874±0.0156
Proposed	**0.9643±0.0014**	**0.7314±0.0114**	**0.8005±0.0061**	**0.8200±0.0054**



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j)

Fig. 10. Change map results of different methods on the River dataset. (a) CVA. (b) SVM. (c) GETNET. (d) WCRN. (e) ChangeFormer. (f) TFR. (g) SST-Former. (h) Without CHM. (i) Proposed CHM-based TWBSIT. (j) Ground truth.

The next model is WCRN, showing an OA of 0.9310, CA of 0.5648, and a Kappa of 0.6637. However, two transformer-based models, ChangeFormer and SST-Former, present similarly weak performances on OA, which are 0.9153 and 0.9061, respectively. The performance results make them the two least-performing deep learning models. As for the traditional methods, CVA and SVM exhibit similar OA performances, which are 0.928 and 0.9147, respectively. Note that the performances obtained by traditional methods are worse in general, which indicate the traditional machine learning methods face difficulty in effectively capturing subtle differences between water and land.

The predicted CD maps of different methods on the River dataset are presented in Fig. 10. In terms of visual representation, the TWBSIT, based on Conflict HyperMatch, is closely consistent with the ground-truth map, demonstrating superior performance. The main differences in every CD map are highlighted with red box. The first CD map is generated by CVA, and it contains numerous scattered areas in the bottom-left corner that do not correspond with the ground truth. For the other traditional method, SVM mistakenly detects scatter points-like changes in the red box area, leading to the bad performance of OA. For deep learning methods, GETNET exhibits a similar visual effect but erroneously identifies edge information in the bottom-left red box mark. The change areas detected by WCRN resemble plaque shapes from visually overall effects, resulting in the loss of much detailed information. When it comes to the transformer architecture-based methods, ChangeFormer generates a result with limited depth, failing to capture intricate change details effectively. TFR exhibits visual effects closely resembling the ground truth, with only minor omissions compared with the proposed method. SST-Former, which achieves the worst performance among deep learning methods, detects much wrong change information in the red box area.

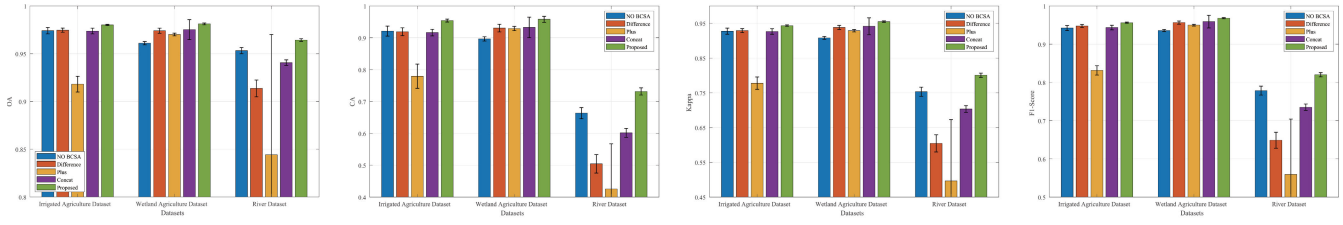
E. Module Ablation

The comparison results of module ablation experiments on the irrigated agricultural dataset can be inspected in the first bar group of Fig. 11(a)–(d). We conduct module ablation experiments by removing BCSA and replacing WITA module with other popular temporal operators. The TWBSIT without BCSA shows a worse performance of OA, which is 0.9741. The results by other different operators show that the addition operator cannot effectively connect temporal information and only achieve an OA of 0.9181. The visual effects of them are shown in Fig. 12. The primary differences are marked with red boxes. The visual effects from difference and concatenation operators are similar, in which only some subtle information are omitted. With reference to the change map obtained by addition operator, many stripe noises are detected mistakenly which influence the OA, finally.

The comparison results of module ablation experiments on the Wetland Agricultural dataset can be inspected in the second bar group of Fig. 11(a)–(d). The ablation experiment results show the effectiveness of our proposed BCSA module and WITA module. From the OA results, the model performs terribly when we remove the BCSA module which decreases to 0.9610. With reference to WITA module ablation, among these popular temporal operators, the addition operator performs the worst OA result which is 0.9701. This indicates that addition operator cannot effectively combine temporal features on the Wetland Agricultural dataset. The visual effects of ablation models are shown in Fig. 13. The main detection differences are marked with red boxes. The visual effects by TWBSIT without BCSA are far from ground-truth map. Among three temporal operators, the addition operator achieves the worst visual effects, as well.

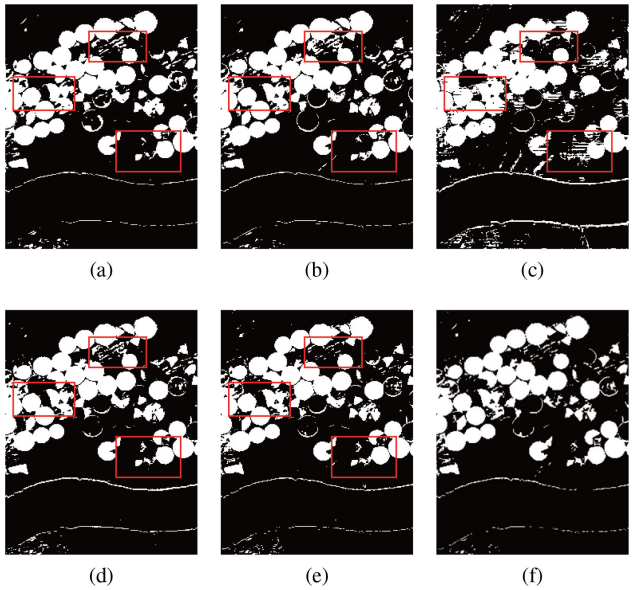
The comparison results of module ablation experiments on the River dataset can be inspected in the last bar group of Fig. 11(a)–(d). When we remove the BCSA module, the

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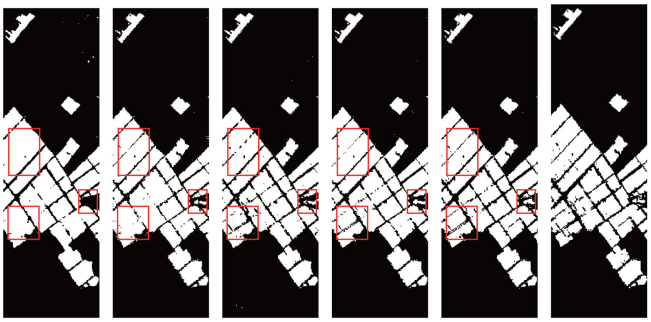
(a) (b) (c) (d)

Fig. 11. Module ablation study results on three datasets (repeat five times), including TWBSIT without BCSA module, WITA module replaced with difference operator, addition operator, and concatenation operator. (a) OA, (b) CA, (c) kappa, and (d) $F1$-Score.



(a) (b) (c) (d) (e) (f)

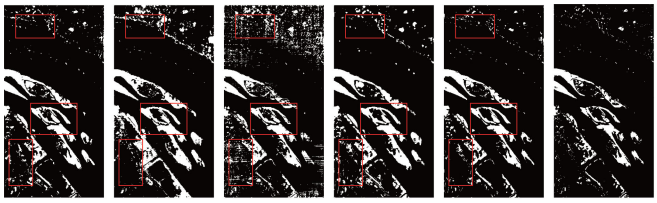
Fig. 12. Change map results of different ablation operators on the Irrigated Agricultural dataset. (a) BCSA removed. (b) Difference operator. (c) Addition operator. (d) Concatenation operator. (e) Proposed model. (f) Ground truth.



(a) (b) (c) (d) (e) (f)

Fig. 13. Change map results of different ablation operators on the Wetland Agricultural dataset. (a) BCSA removed. (b) Difference operator. (c) Addition operator. (d) Concatenation operator. (e) Proposed model. (f) Ground truth.

model performance decreases to 0.9532, compared with the proposed method. The ablation results obtained by popular temporal operators prove that our proposed WITA module can more effectively leverage temporal interaction information. Note that the addition operator still performs worst which is 0.8441 of OA. The comprehensive visual results are depicted in Fig. 14 and the main differences are marked with red boxes. The visual effects obtained by difference and concatenation operators mainly mistakenly detect subtle change areas in the bottom-left corner. However, addition operator shows the worst visual effects which contains many stripes and scatter



(a) (b) (c) (d) (e) (f)

Fig. 14. Change map results of different ablation operators on the River dataset. (a) BCSA removed. (b) Difference operator. (c) Addition operator. (d) Concatenation operator. (e) Proposed model. (f) Ground truth.

point-like noise. This indicates that addition operator cannot combine the temporal features.

F. Hyperparameters' Analysis

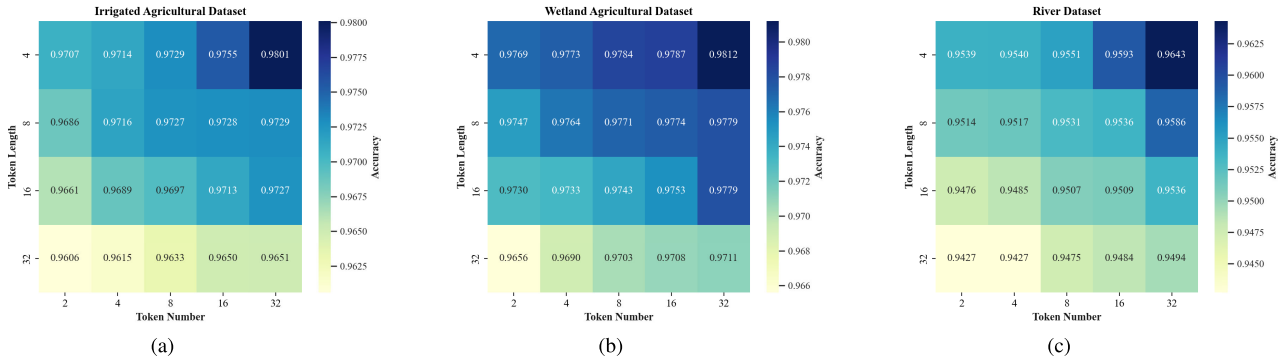
For hyperparameters' analysis, the parameter sets for token, patch size, sample rate effects for the Conflict HyperMatch strategy, and loss item coefficients are chosen as the hyperparameters' ablation analysis. The comprehensive results support our proposed method, and we eventually choose the optimal parameters as the default setting.

1) Ablation for Tokens: The number of tokens and the length of tokens are analyzed in Fig. 15. Fig. 15(a) shows that the detection accuracy increases when we set more token numbers. However, the OA decreases along with the increase in token length. It could result from the redundant projection for spectral features that do not contain much change information. The same trend can be observed from both Fig. 15(a) and (c). Therefore, based on the hyperparameter ablation for tokens, the optimal token number is set as 32 and the token length is set as 4.

2) Hyperparameters' Ablation for Conflict HyperMatch Loss Function: The hyperparameters' ablation for Conflict HyperMatch loss function is also analyzed which includes the coefficient λ_d for discrepancy loss and the coefficient λ_u for unsupervised loss item. The analysis results on three datasets are shown in Fig. 16. From the visual observation, the highest performances are achieved when both λ_d and λ_u are set as 0.2. If the coefficient values increase, the model performances behave worse, which implies that Conflict HyperMatch is valuable for model learning on unlabeled samples. Therefore, we set the default parameter values with the optimal results.

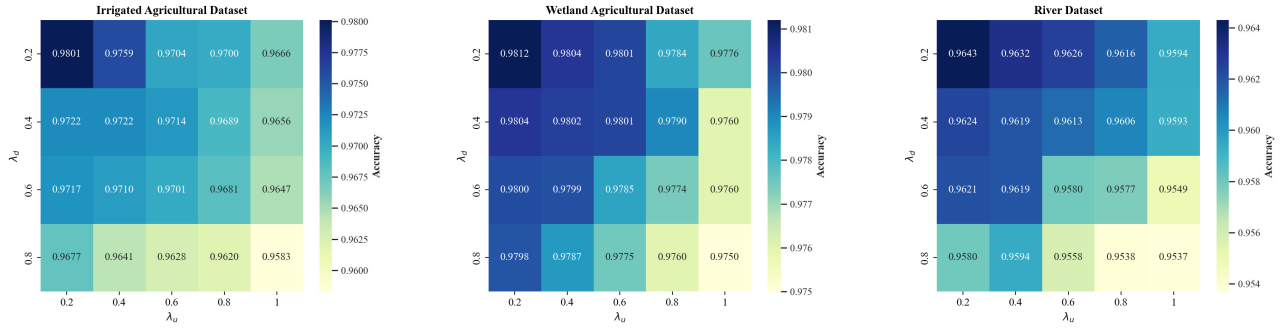
3) Ablation for Patch Size and Training Rate: The patch size analysis and training sample rate analysis are shown in Fig. 17. As Fig. 17(a) shows, the model performance can obtain the best results when the patch size is set as 5×5. Then, the model performance decrease remarkably when the

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(a) (b) (c)

Fig. 15. Token ablation analysis on three datasets. (a) Irrigated Agricultural dataset. (b) Wetland Agricultural dataset. (c) River dataset. Dark colors imply better OA performance and light colors imply worse OA performance.



(a) (b) (c)

Fig. 16. Hyperparameters' ablation for Conflict HyperMatch loss function on three datasets. (a) Irrigated Agricultural dataset. (b) Wetland Agricultural dataset. (c) River dataset. Dark colors imply better OA performance and light colors imply worse OA performance.

patch size gets larger. This could result from that the larger patch size would introduce irrelevant spatial information which is harmful to the texture and boundary information. The training sample rate analyses are shown in Fig. 17(b)–(d), where the red dashed line indicates TWBSIT with Conflict HyperMatch, while the black line indicates regular supervised training for TWBSIT. The asterisk-marked 100% sample rate indicates using the initial supervised benchmark training sample which are 9.7%, 9.7%, and 4.03% of the total samples. Note that the standard benchmark training sample rate is explained in the dataset section, which follows their originally supervised works. Therefore, the training rate analysis is based on the product of the ratio and initial training samples, for example, 40% $\times$ 9.7%. From the visual effects, it can be observed that Conflict HyperMatch can effectively leverage unlabeled samples. As the number of training samples decreases, the effectiveness of Conflict HyperMatch becomes more evident, and the gap in performance between whether to use Conflict HyperMatch widens. Therefore, the proposed Conflict HyperMatch training strategy can evidently decrease the demand for labeled samples and eventually has the potential to relieve the heavy labor cost.

G. Model Efficiency

To compare the efficiency of the test methods, Table V reports the flops and parameters of the different methods. Due to the flops and parameters are algorithm fixed attributes, we choose the River dataset to illustrate the relative OA and Kappa achievements for each method. The proposed TWBSIT model contains multiple BCSA modules, which can use densely bandwise spatial interaction among entire patch size. The proposed model has adopted WITA module and SA module. These modules all need the long-range resemblance computation which inevitably introduces more computation because the amount of information in an HSI is much larger than that in text. However, the experimental results show that the model performance is substantially improved. For example, the number of parameters of TWBSIT is higher than that of the convolution-based GETNET and WCRN, but the OA of TWBSIT is improved by around 0.03 compared with GETNET. While compared with other transformer-based method, our method shows better performance while keeping less parameters. For instance, the number of parameters of TWBSIT is less than that of ChangeFormer while the OA of TWBSIT is improved by around 0.05 compared with ChangeFormer. This illustrates the effectiveness of the proposed method to capture sharing and bandwise interaction relationship from the temporal, spatial, and spectral domains.

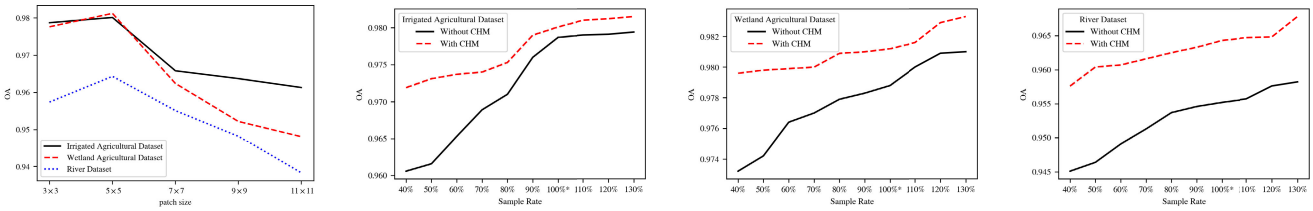
TABLE V
MODEL EFFICIENCY ANALYSIS ON RIVER DATASET

Method	Flops	Param	OA	Kappa
GETNET	0.17 G	0.16 M	0.9352	0.6821
WCRN	0.31 G	0.13 M	0.9310	0.6637
ChangeFormer	181.51 G	29.82 M	0.9153	0.5519
TFR-PS^{2}ANet	53.82 G	12.77 M	0.9488	0.7340
SSTFormer	9.17 G	2.48 M	0.9061	0.5929
Proposed	15.80 G	4.04 M	0.9643	0.8005

V. DISCUSSION

The extensive comparison and ablation experiments illustrate the superiority of our proposed semi-supervised

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(a) (b) (c) (d)

Fig. 17. Patch size ablation analysis and training sample rate ablation analysis. (a) Patch size. (b) Training rate on the Irrigated Agricultural dataset. (c) Training rate on the Wetland Agricultural dataset. (d) Training rate on the River dataset.

Conflict-HyperMatch-based TWBSIT crossing metrics of OA, CA, Kappa, and $F1$-score. For comparison of conventional methods, both unsupervised CVA and supervised SVM cannot harness the abundant temporal, spatial, and spectral information, which eventually cause inferior detection results. For the deep learning methods, convolution-based GETNET and WCRN are designed to extract local spatial–spectral features which cannot use temporal resemblance. The transformer-based ChangeFormer, TFR, and SST-Former can leverage local temporal–spatial–spectral dependencies. However, there are shortcomings in aspects of capturing sharing temporal interaction and intensively long-range spatial resemblances.

The proposed Conflict HyperMatch is aimed to leverage both the labeled and unlabeled samples. Especially for unlabeled samples, the proposed training strategy can encourage deep model to reason weak-to-strong consistent multidimensional predictions while distinguishing perturbations from views with different augmentations. In real-world situations, the labeling process is typically labor-intensive, particularly during on-site investigations and visual interpretations for hyperspectral CD. Our proposed method can preserve high CD accuracy when the labeled training samples significantly decrease, which indicates that the proposed Conflict HyperMatch can effectively reduce the demand for labeled samples.

However, there are some limitations of the proposed method. For example, extra time cost is needed due to the two-phase training schedule. The Conflict HyperMatch is set as a pretext task that can leverage both a small set of labeled samples and a large number of unlabeled samples and this would inevitably spend more time in full training. Furthermore, the semantic features of land-cover temporal–spatial–spectral information are still underused. Abundant spectral changes according to the real land type can be clustered into a more various change classification paradigm which would increase semantic interpretation. To handle these challenges, a potential approach can be leveraging memory-efficient training strategies and developing change-type interpretation module to restrain feature distribution in advance.

VI. Conclusion

This article presents a semi-supervised training strategy and a deep learning model, called Conflict HyperMatch and TWBSIT, respectively. The Conflict HyperMatch uses both the representation discrepancy and the weak-to-strong consistency approaches to overcome the challenge of limited labeled samples. First, both a few of labeled samples and a large amount of unlabeled samples are perturbated differently. The weakly perturbated labeled samples are trained to keep consistency to ground truth, while weakly and strongly perturbated unlabeled samples are aimed to give consistent predictions while distinguishing perturbated representation from views with different augmentations. Second, a few of real labeled samples are used to fine-tune the TWBSIT model in a supervised manner. For the proposed deep learning model, TWBSIT is aimed to capture weight-sharing attention map and long-range bandwise spatial resemblance for better and more effective temporal–spatial interaction. Therefore, the WITA module and BCSA module are developed.

The proposed method is executed on three publicly available hyperspectral CD datasets. The results show that our proposed method can achieve both visual and quantitative effects, which demonstrate the superior performance when compared with other SOTA methods.

For the value in diverse CD tasks, the Conflict-HyperMatch-based TWBSIT can be regarded as a benchmark model with an attention mechanism and can serve as a reference for other semi-supervised strategies, where the pretext task is without manual intervention. For future research, we will introduce semantic explanation potential into our model to explore its effectiveness based on prior knowledge of general land change types.

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